MIL-M-748D
23 June 1978
SUPERSEDING
MIL-M-748C
10 August 1970

MILITARY SPECIFICATION

MERINGUE, POWDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a meringue powder which can be readily dispersed by the addition of water, and whipped to provide a meringue topping for bakery, desserts and confection items. The product is intended for use by the Armed Forces as an item of general issue.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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ASSOCIATION OF THE OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official nalytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, C 20044.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. All materials shall be clean, sound, and wholesome, and free from evidence of insect infestation or any other objectionable foreign matter, foreign flavor or odor.

3.1.1 Sugar, powdered. Sugar shall be white, refined cane or beet sugar, or a combination of both, and shall be of the granulation known as powdered.

3.1.2 Cornstarch. Cornstarch shall be white in color, and possess a bland flavor and odor.

3.1.3 Egg albumen, dehydrated. Dehydrated egg albumen shall have been derived from grade C or better, shell eggs or from frozen shell egg albumen manufactured in compliance with US Department of Agriculture publication entitled: "Regulations Governing the Grading and Inspection of Egg Products". Dehydrated egg albumen shall be prepared under continuous inspection of the USDA and shall bear the inspection shield mark. Dehydrated egg albumen shall have been laboratory tested and certified Salmonella Negative by the US Department of Agriculture. The dehydrated egg albumen shall not exceed 8.0 percent moisture, and the reducing sugar content shall not exceed 0.1 percent on a moisture free basis.

3.1.4 Flavoring materials. Flavoring material may be either natural or artificial or a combination of both.

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3.1.5 Monocalcium phosphate. Monocalcium phosphate shall be a white, crystalline, free-flowing powder.

3.1.6 Sodium aluminum sulfate. Sodium aluminum sulfate shall be a white, crystalline, free-flowing powder.

3.1.7 Stabilizers. Stabilizers may include gums, soluble starches or any other material which will meet the performance requirements specified in 3.3.1.

3.1.8 Salt. Salt shall be white, refined, sodium chloride, with or without anticaking agents. Iodized salt shall not be used.

3.1.9 Dextrose. Dextrose shall be white, anhydrous, and pulverized.

3.2 Material.

3.2.1 Formulation. The meringue powder shall contain dehydrated egg albumen and powdered sugar. Optionally, the manufacturer may add one or more of the following ingredients: Cornstarch, dextrose, monocalcium phosphate, sodium aluminum sulfate, flavoring materials, stabilizer and salt, to the extent necessary to meet the performance requirements specified in 3.3.1.

3.3 Finished product. The finished meringue powder shall be of uniform composition. It shall have uniform color ranging from not lighter than FED-STD-595 color chips 37875 to not darker than 37855. (Note: Color chips may be obtained from General Services Administration Business Service Center, Region #3, Washington, DC 20407.) Meringue powder and chips shall be compared under daylight conditions and held in such a way as to avoid specular refractance (glassiness). The moisture content of the product shall not exceed 7.0 percent. It shall be free from foreign material or foreign odors, e.g., stale, putrid, rancid, etc., and it shall not contain any lumps that do not break up under light finger pressure. The product shall comply with 3.3.1.

3.3.1 Performance requirements. Meringue powder (prepared according to 4.4.2.2) shall yield a product that approximates the sensory qualities of a meringue made from fresh eggs (prepared according to 4.4.2.1). The test meringue shall meet the following requirements:

(a) Texture, odor, and color of unbaked test meringue shall be approximately the same as these qualities in the unbaked control meringue.

(b) One quart (.946 liter) of freshly prepared (see 4.4.2.2), unbaked test meringue shall weigh not more than 9 avoirdupois ounces (255 g).

(c) When 1 quart (.946 liter) of unbaked test meringue is stored in a refrigerator (35° to 40°F) (1.67° to 4.44°C) for 4 hours, not more than 20 milliliters (0.676 liquid ounces) of fluid shall separate.

(d) Texture, odor, flavor and color of baked test meringue shall be approximately equal to baked control meringue.

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(e) When baked test meringue is stored at room temperature (60° to 80°F) (15.56° - 26.67°C) for 5 hours after removal from the oven and the height of the meringue in the center of the layer is measured, the height shall be not less than that of the unbaked test meringue (minus 1/4 inch) (6.35 mm).

(f) Baked test meringue shall maintain its form for at least 16 hours after removal from the oven when stored at room temperature (60° to 80°F) (15.56° to 26.67°C) and within an atmospheric relative humidity of 30 to 75 percent.

(g) Not more than two cracks each measuring more than 1 inch (25.4 mm) in length shall appear in the surface of the baked test meringue.

3.4 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.5 Workmanship. The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection. Product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.3 Examination of end item. Examinations shall be in accordance with tables III and IV and 4.3.3.1 through 4.3.3.5. Sampling plans for the tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as percent defective for table III and defects per hundred units for table IV.

TABLE II. Sampling plans

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
III	S-3	One filled can	Cans	-	2.5
IV	S-2	One filled can	Cans	1.5	-

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Less than 15 1/2 ounces (0.44 kg)

* 1/ Report results to a tolerance of 1/8 ounce (3.5 grams). Lot shall be rejected if sample data indicate a lot average net weight less than specified.

TABLE IV. Examination of product characteristics 1/

Category	Examine	Defect
Major		
101	Powder	Not uniform composition.
102		Color lighter than 37875 or darker than 37855 of FED-STD-595.
103		Presence of lumps that do not break up under light finger tip pressure.

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TABLE IV. Examination of product characteristics 1/ (cont'd)

Category	Examine	Defect
Major		
104	Powder (cont'd)	Objectionable odor, e.g., stale, putrid, or rancid.
	Unbaked whipped meringue (see 4.4.2.2 for preparation - Prepare one per sample unit)	
105		One quart (0.95 liters) of meringue weighs more than 9 ounces (255 g).
106		More than 20 ml (0.68 liq oz) of fluid separates from 1 quart (0.95 liters) of meringue after 4 hours of storage between 35° to 40°F (1.67 - 4.44°C).
	Baked meringue (see 4.4.2.1 and 4.4.2.2 for preparation - Prepare one per sample unit).	
107		Not approximately equal in texture, odor and color to baked control meringue.
108		More than two cracks that measure more than 1 inch (25.4 mm) long on surface of meringue.
109		Height in center of meringue more than 1/4 inch (6.35 mm) lower than unbaked test meringue after 16 hours of storage between 60° and 80°F (15.56 - 26.67°C) and within a relative humidity of 30 - 75 percent.

1/ Presence of foreign material, e.g., wood, metal, glass, insects, insect fragments, filth, etc., shall be cause for rejection of the lot.

* 4.3.3.1 Examination of cans. Examination of cans for external condition shall be in accordance with the United States Standards for Condition of Food Containers. Examination for seam leakage shall be in accordance with the applicable quality assurance provision of PPP-C-29.

4.3.3.2 Examination of can coating. Examination of can coating (when required) shall be in accordance with the quality assurance provisions of TT-C-495.

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4.3.3.3 Examination of can label. Examination of can label shall be in accordance with the quality assurance provisions of MIL-L-1497.

* 4.3.3.4 Examination of shipping container. Examination of shipping container shall be in accordance with the quality assurance provisions in the appendix of PPP-B-636. In addition the following defects shall be included in the examination: Major - NSN, Item description, contract No., Date of pack missing, incomplete, incorrect, or illegible; Minor - (1) Other specified markings missing, incomplete, incorrect, or illegible; (2) Arrangement or number of cans not as specified.

4.3.3.5 Examination of unit loads. Unit loads shall be examined in accordance with the criteria in MIL-L-35078.

* 4.3.4 Testing of finished product. The finished product shall be tested for moisture content in accordance with 4.4. The sample for testing shall be a 4 ounce (0.11 kg) composite derived from the number of cans indicated by inspection level S-2. Lot size shall be expressed in cans. Result shall be reported to the nearest 0.1 percent. Nonconformances to the test requirements shall be cause for rejection of the lot.

4.4 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the contractor will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.4.1 Test procedures.

4.4.1.1 Moisture. Test for moisture in the finished product shall be made in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Cereal Foods, Section: Wheat Flour, Method: Total Solids, Vacuum Oven.

4.4.1.2 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.4.2 Examination procedures.

4.4.2.1 Preparation of control meringue. (No alternate procedure permitted.) Control meringue shall be prepared as follows: Place 13 avoirdupois ounces (368.5 g) of fresh egg whites from grade A or better eggs having an initial temperature of 70°F (21.11°C) in the 10-quart (9.46 liter) bowl of a Hobart C-100 electric mixer (or equivalent) and beat at high speed until stiff peaks

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are formed. Add 1 pound (0.45 kg) of fine granulated sugar in two equal portions, mixing until the sugar is dissolved after addition and whipping for 7 minutes on second speed after the second addition. Spread one 1-3/4-inch (44.45 mm) layer of control meringue on the bottom of 1 inverted 8-inch (203,2 mm) diameter circular aluminum cake pan, tapering the sides of the meringue at a 45 degree angle toward the center of the top of the layer, and bake in a hot oven (350° ± 10°F) (70° ± 12.22°C) for 16-20 minutes.

4.4.2.2 Preparation of test meringue. (No alternate procedure permitted.) Test meringue shall be prepared as follows: Add 3 avoirdupois ounces (85 g) of meringue powder to 16 avoirdupois ounces (453.6 g) of water at 70°F (21.11°C) and mix in the 10-quart (9.46 liter) bowl of a Hobart C-100 electric mixer (or equivalent) until stiff peaks are formed. This stage should be reached in not more than 10 minutes of mixing. Add 1 pound (0.45 kg) of fine granulated sugar in two equal portions, beating after each addition until sugar is dissolved. Spread and bake one layer of test meringue in the same manner as specified in 4.4.2.1.

5. PACKAGING

* 5.1 Preservation-packaging. A net weight of one pound (0.45 kg) of the product shall be packaged in accordance with level A or C, as specified (see 6.1). A minus 1/2 ounce (14 g) tolerance will be allowed in any one can provided that the average net weight of the cans inspected in accordance with table III, is not less than the stated net weight.

* 5.1.1 Level A. The product shall be filled into a size 401 by 411, open-top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. The cans shall be made throughout from not less than commercial 0.25-pound per base box electrolytic tin plate (see 4.4.1.3) and shall be coated outside with a coating conforming to type I or II, or when specified (see 6.1), type III of TT-C-495. The cans shall be hermetically sealed.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing. Twenty-four metal cans of the product, arranged four in length, three in width, and two in depth, shall be packed on end in snug-fitting shipping containers in accordance with level A, B or C as specified (see 6.1).

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5.2.1 Level A. The container shall be a fiberboard box constructed and closed in accordance with style RSC, V2s of PPP-B-636.

5.2.2 Level B. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V3c, V3s, or V4s, of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.2), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. The product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1).

5.4 Labeling and marking.

5.4.1 Metal cans. Metal cans shall be labeled in accordance with MIL-L-1497. In addition, the following shall appear on the body of the can:

MERINGUE POWDER (in letters larger than any others used for the label)

_____ Net weight

Name and address of manufacturer

_____ Date packed (month and year) (coding not permitted)

Any other information required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder

In addition, the following recipe shall be included with the labeling information:

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Recipe for Meringue Topping

YIELD: 2-1/2 gallons, approximately 2-1/4 to 2-1/2 cups per pie)

Ingredients	Weights	Measures	Method
Meringue powder	6 oz. (170.1 g)	1-1/2 cups	1. Add meringue powder to water in mixer bowl. Using whip, mix 1 minute at low speed; scrape down bowl; change to high speed and beat 5 minutes or until mixture forms stiff peaks.
Water, cold		1 qt. (0.95 liters)	
Sugar, granulated	2 lb. (0.91 kg)	4-1/2 cups	2. Gradually add sugar to mixture while beating at medium speed. Continue to beat 1 minute at high speed or until meringue is glossy.
			3. Spread meringue over warm filling (122°F) (50.4°C) to the edge of the crust so the top of each pie is completely covered. Leave meringue somewhat rough on top.
			4. Bake in a 350°F (70°C) oven 16-20 minutes.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required (see 5.1 and 5.2).
(c) When type III exterior can coating is required (see 5.1.1).
(d) When strapping specified in 5.2.2.1 is required.
(e) Type and class of unit load required (see 5.3).

6.2 Packing specified in 5.2.2.1 is intended for transfer at sea operations; and specific overseas operations.

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6.3 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:

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Navy - MC, MS
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Preparing activity:

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-748D, Meringue, Powder

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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